Atomic Theory

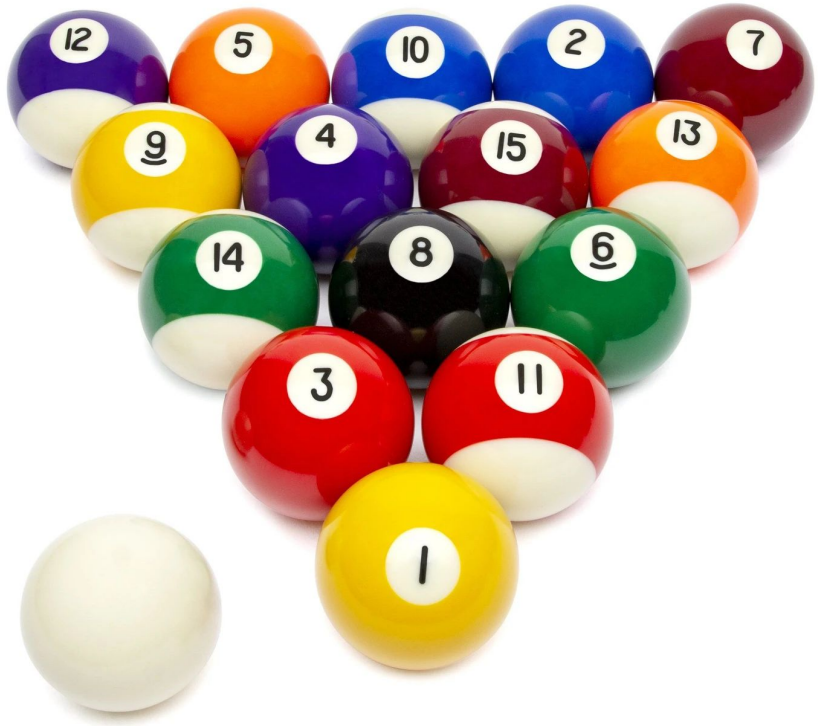
MODELS OF THE ATOM

Ancient Greece

- All matter is composed of small particles that they called ***atomos***, a term derived from the Greek word for "**indivisible**"
- They thought of atoms as moving particles that differed in **shape** and **size**, and which could join together

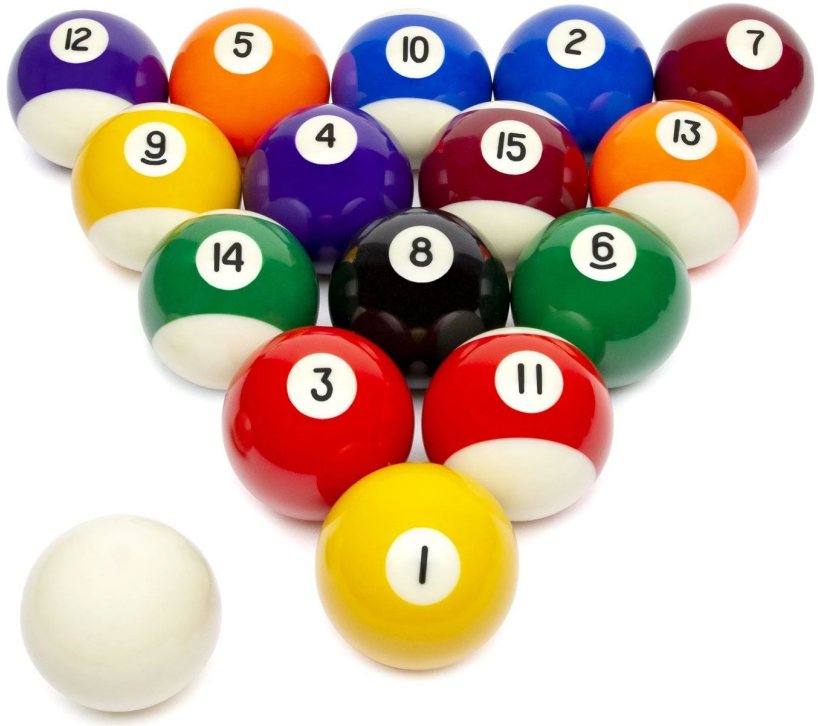
The Billiard Ball Model (1803)

- All matter is made up of tiny, indivisible particles called **atoms**
- All atoms of an element are **identical**
- Atoms of different elements are **different**



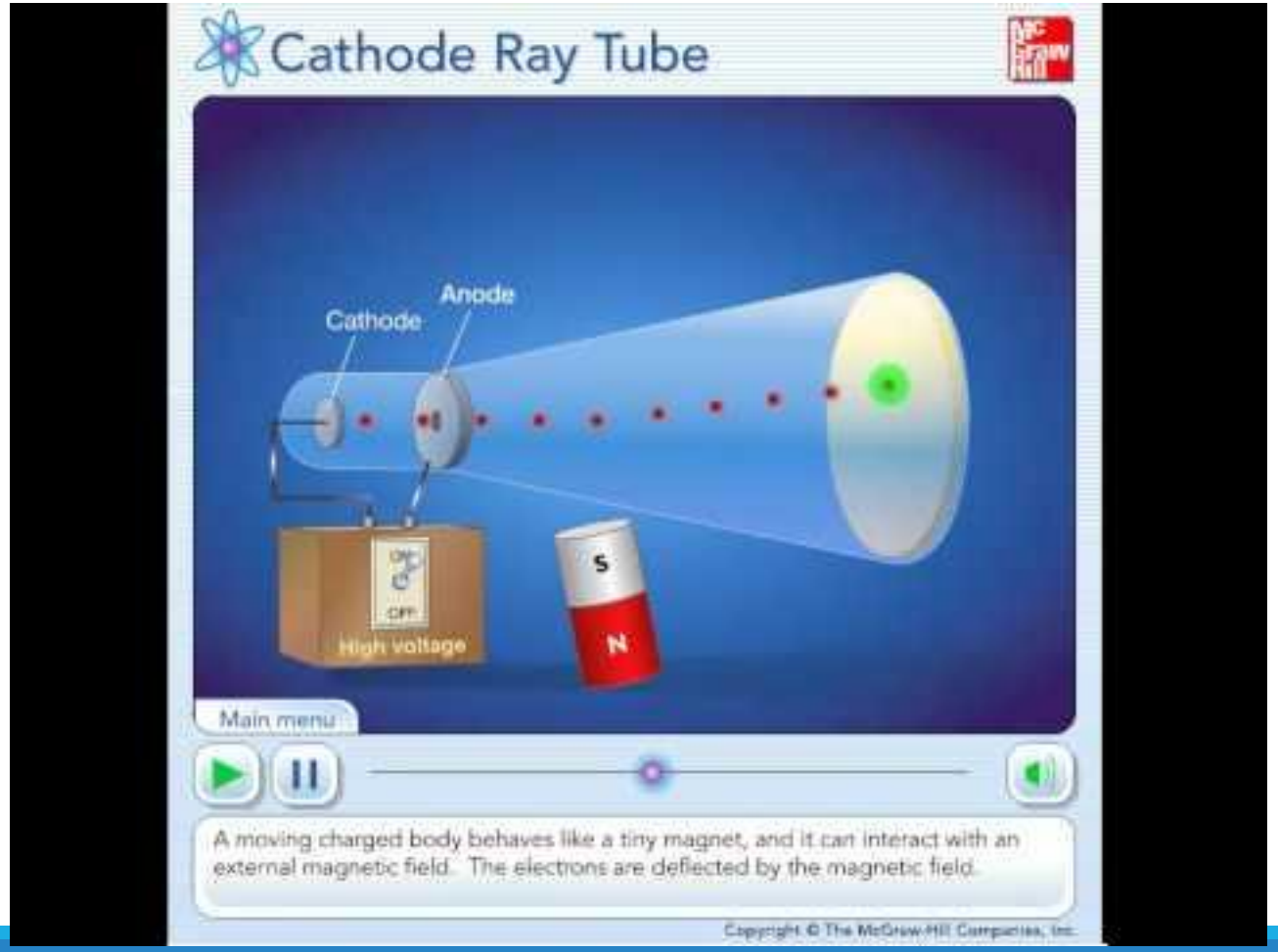
Limitations of the Billiard Ball Model

1. It could not explain why different elements had different **masses**
2. It could not explain many **experimental** observations, particularly those involving gases



Discovery of the Electron (1897)

- During the 1880s and 1890s scientists searched cathode rays for the carrier of the **electrical** properties in matter
- This led to the discovery of the **electron**

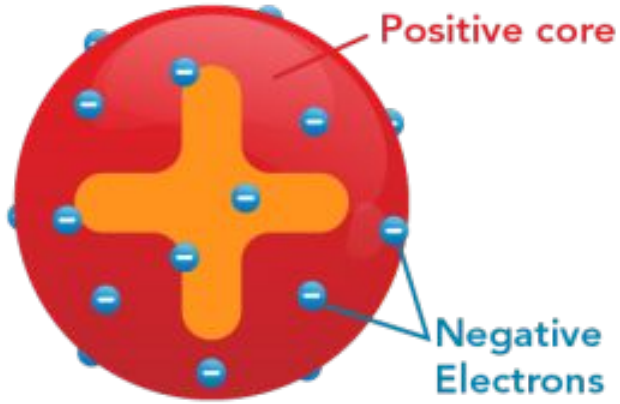


Plum Pudding Model (1904)

- Atoms are composed of **negatively** charged **electrons** scattered throughout a spherical cloud of **positive** charge

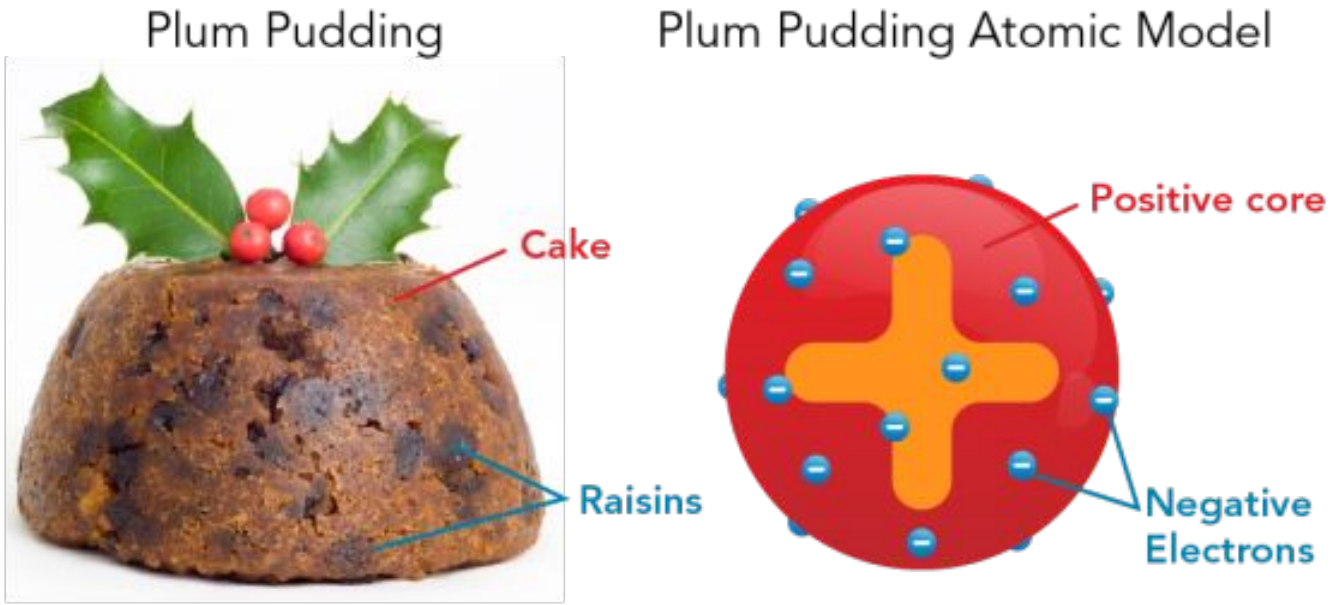
Plum Pudding

Plum Pudding Atomic Model



Plum Pudding Model (1904)

- An atom as a whole is electrically **neutral** because the negative and positive charges are **equal** in size

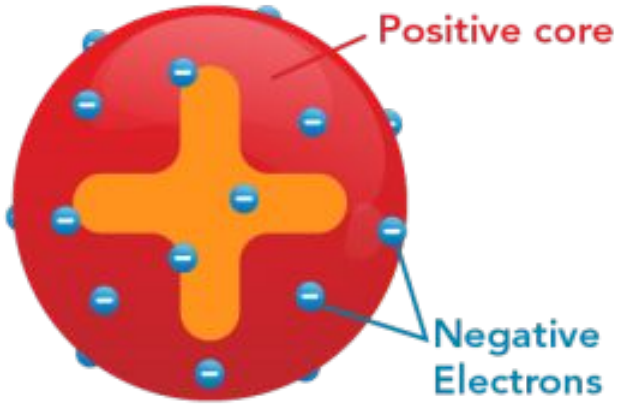


Limitations of the Plum Pudding Model

- There was **no** experimental **evidence** to support the model

Plum Pudding

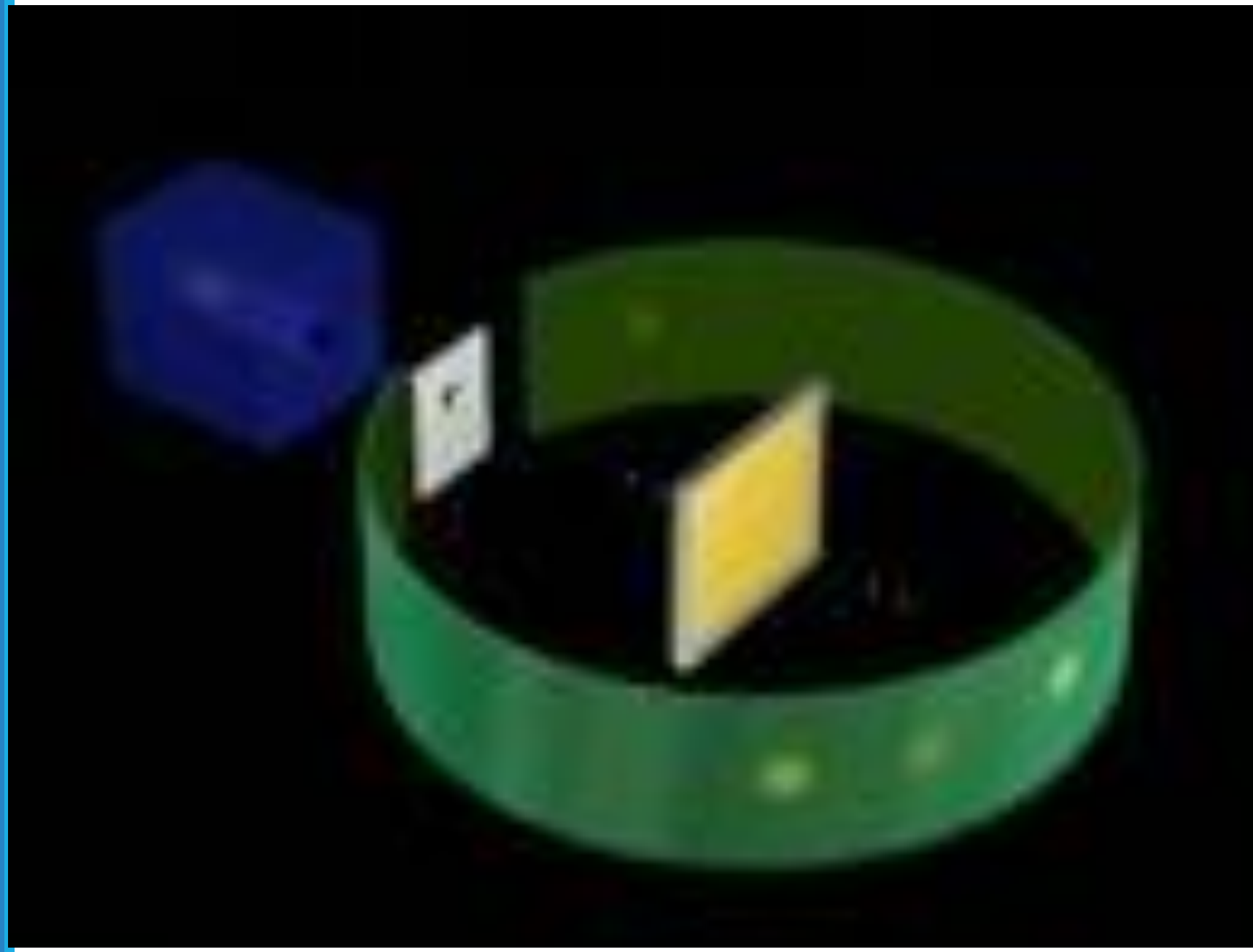
Plum Pudding Atomic Model



The Gold Foil Experiment (1910)

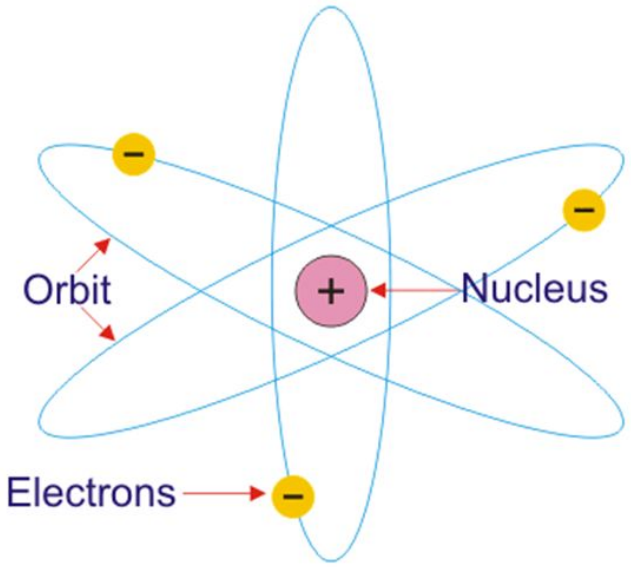
- Rutherford shot tiny, positively charged particles called **alpha** particles at a thin piece of gold foil
- Observed that many of the particles passed **through** the foil unaffected, but some were **deflected** at very large angles, as though something very massive yet very small were repelling them

The Gold Foil Experiment (1910)



The Nuclear Model (1911)

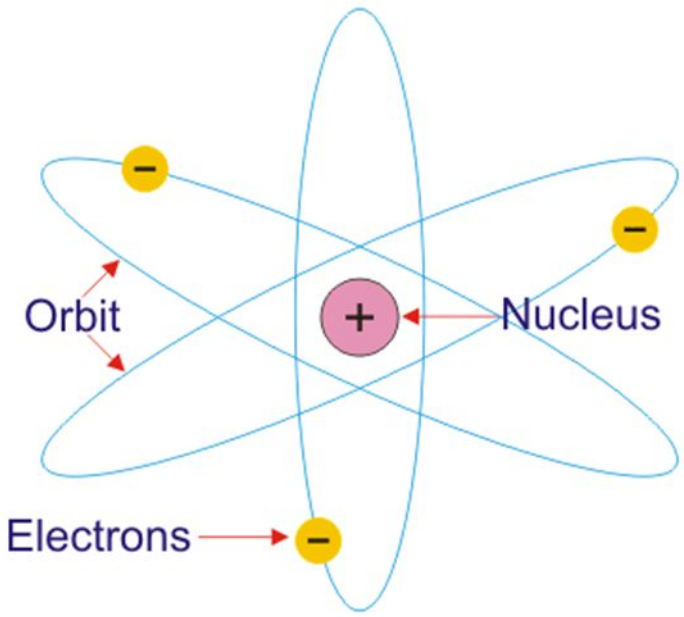
- The center of the atom has a **positive** charge, which we call the **nucleus**
- The nucleus contains most of the atom's **mass** but occupies a very small **space**
- The nucleus is surrounded by a cloud of negatively charged **electrons**
- Most of the atom is empty **space**



Rutherford's Atomic Model

Limitations of the Nuclear Model

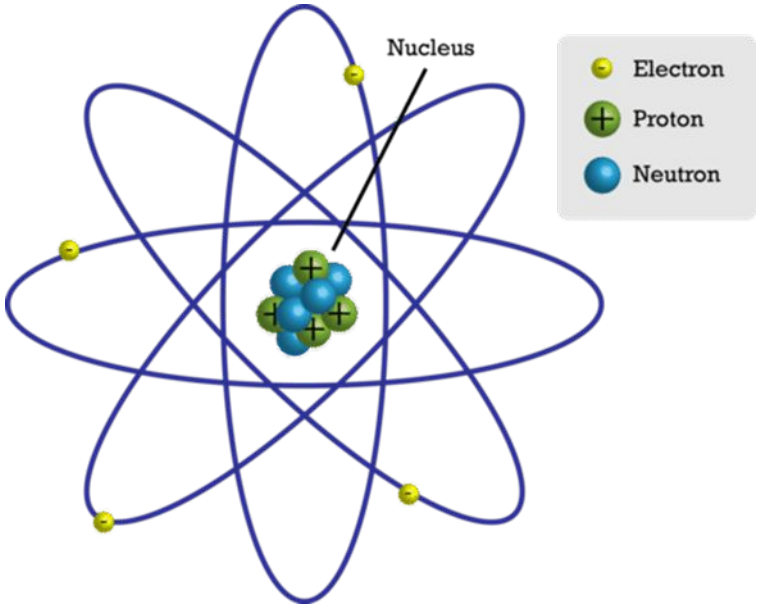
1. If the nucleus is entirely composed of positive charges, shouldn't it break apart due to electrostatic **repulsions** ?
2. The model couldn't account for the total **mass** of the atom



Rutherford's Atomic Model

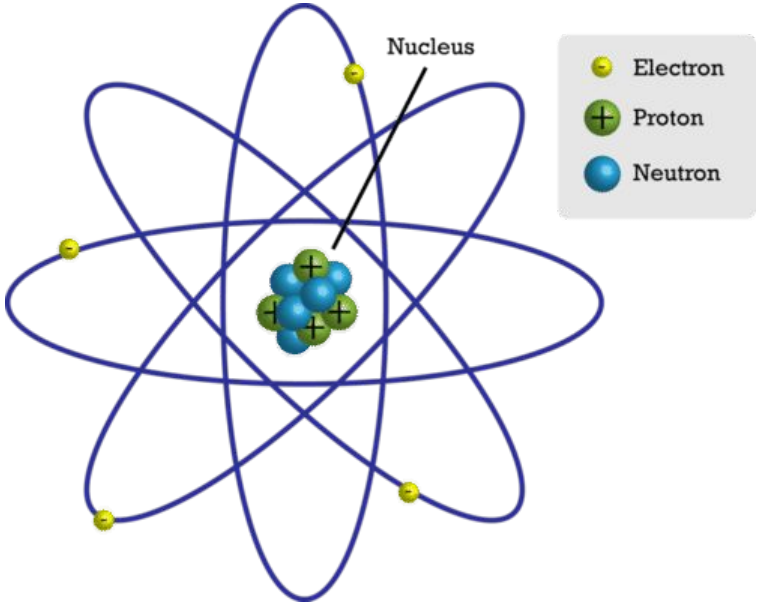
Discovery of Protons & Neutrons

- A series of **experiments** by Rutherford led to the discovery of the **proton** in 1919
- A **neutral** particle, called the **neutron** was discovered in 1932
- The Nuclear Model was then **revised**



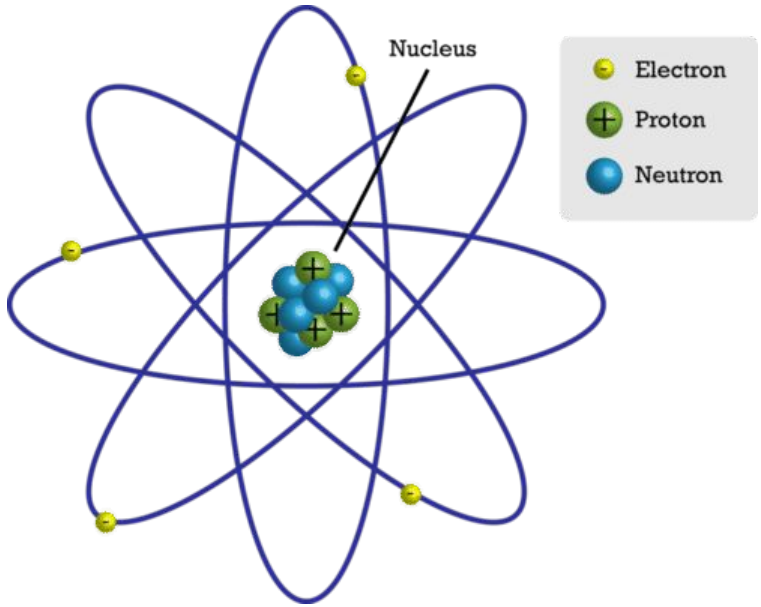
The Revised Nuclear Model

- An atom is an empty sphere with a tiny **dense** central **nucleus**
- This nucleus contains positively charged **protons** and neutral particles called **neutrons**
- The **mass** of a neutron is about the **same** as a proton



The Revised Nuclear Model

- Negatively charged **electrons** circle rapidly through empty space around the nucleus
- There are **equal** numbers of **electrons** and **protons** in a <u>neutral</u> atom



Your turn

- Atomic Theory Summary